

Antibacterial Activities of Bacteriagenic Silver Nanoparticles Against Nosocomial *Acinetobacter baumannii*

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Acinetobacter baumannii has emerged as one of the major nosocomial pathogens implicated in variety of severe infections and mortality. It is rapidly developing multi-drug resistance and also possesses surface colonization ability, which make it most difficult to treat through traditional antibiotics. This is an extensive study to describe the antibacterial activity of bacteriagenic silver nanoparticles (AgNPs) against *A. baumannii* ATCC 35061 in planktonic and biofilm mode. Minimum inhibitory concentration of antibiotics were in the range of 1 to 4096 µg/ml whereas AgNPs inhibited planktonic bacteria at concentration of 16 µg/ml. Fractional inhibitory concentration index revealed the synergistic interaction of AgNPs with doxycycline, tetracycline and erythromycin. Nanoparticles exhibited significant biofilm disruption activity with minimum biofilm eradication concentration of 2 mg/ml. Eradication of mature biofilm was enhanced on exposure to combination of AgNPs and antibiotics. These nanoparticles affected bacterial growth and distorted cellular morphology. Intracellular oxidative stress, induced in presence of AgNPs, also rendered bacteria susceptible to killing by nanoparticles. Besides this, AgNPs were found to interact with thiol-groups, which indicate their potential to interact with cellular proteins to exhibit antimicrobial activity.

Keywords: Silver Nanoparticles (AgNPs), *Acinetobacter*, Synergy, Biofilm, Mechanism.

1. INTRODUCTION

Acinetobacter species are Gram negative, non motile, aerobic and coccobacillary bacteria.¹ The genus consists of 23 species with formal nomenclature. Moreover, there are at least 11 additionally recognized species, belonging to this genus, without name.^{2,3} These are ubiquitous in nature and widely distributed in soil, water, food, industrial waste, activated sludge and human skin, including mucus membrane of upper respiratory tract.⁴⁻⁷ However, they have emerged as the most important nosocomial pathogens in recent years. *A. baumannii* has been implicated in hospital-acquired infections, such as bacteremia, skin and soft-tissue infections, urinary and respiratory tract infections and secondary meningitis leading to morbidity and mortality in patients.^{8,9} Besides this, the bacteria have also been associated with chronic injury and burn wound infection,

ulcers and combat casualties in military people.^{10,11} However, the problem is aggravated owing to two reasons— (a) development of multi-drug resistant (MDR) strains in bacteria, which render the antibiotics inefficient to treat the *Acinetobacter* infections, and (b) ability to form biofilm on biotic and abiotic surfaces, which significantly enhances the resistance of bacteria to antibiotics due to protective barrier of extracellular polymeric substances (EPS).¹²⁻¹⁴ Moreover, bacterial cells in biofilm mode express different phenotypes, as compared to their planktonic counterparts, and exhibit higher resistance to extreme pH, desiccation and host's immune defense system.¹⁵⁻²⁰ Because *A. baumannii* can evade the biocidal activity of antibiotics, it has been enlisted as one of the ESKAPE pathogens.²¹ In view of these facts, it is necessary for us to devise novel alternative strategies to combat MDR *Acinetobacter* infections, one of which could be to employ nanomaterials.

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Nanoparticles have gained significant importance in a decade and have been applied in various fields of medicine, therapeutics and diagnosis.^{22–27} Metal nanoparticles possess high specificity, selectivity and sensitivity due to large surface area to volume ratio, and have unique physical, chemical, mechanical, optical, electrical, thermal, magnetic and biological properties owing to their nano size.^{28,29} Silver nanoparticles (AgNPs) of size <100 nm are reported to have excellent antimicrobial activity against Gram positive and Gram negative bacteria, mycobacteria, yeasts and fungal pathogens.^{28,30–32} They also exhibited antibiofilm activity.^{1,25,33,34} AgNPs have been shown to disrupt the cellular morphology and functioning to exert antimicrobial effect.^{35–37} These nanoparticles can be synthesized by physical and chemical routes.^{38–40} Biological methods involve synthesis using bacteria, fungi, actinomycetes and plants.^{6,23,30,41} Bio-nanoparticles are often regarded safe, non-toxic and biocompatible as compared to chemically synthesized nanoparticles, which employ harsh chemicals and produce toxic by-products.⁴²

In our previous study, we reported the synthesis of spherical AgNPs *Acinetobacter calcoaceticus* LRVP 54 isolated from rhizosphere of wheat.⁴³ In present study, we compared the efficacy of these bacteriogenic AgNPs and antibiotics to inhibit *A. baumannii* in planktonic and biofilm mode. Nature of interaction between AgNPs and antibiotics in combination has also been established. Besides this, few assays were carried to predict the mechanistic aspects of antimicrobial activity of these nanoparticles.

2. EXPERIMENTAL DETAILS

2.1. Antibiotics and AgNPs

Antibiotics, namely, amoxicillin, ampicillin, doxycycline, tetracycline and erythromycin (HiMedia, Mumbai, India) were used for antimicrobial assay. AgNPs were synthesized by 72 h cell-free extract of *A. calcoaceticus* LRVP 54, isolated from wheat rhizosphere, as per the protocol described in our earlier study.⁴³ Transmission electron microscope revealed the formation of spherical monodispersed AgNPs with size 8–12 nm.⁴³

2.2. Culture

A. baumannii AIIMS 7 [GenBank EU779829] isolated and identified in our laboratory has been used to investigate the antibacterial and antibiofilm activity of AgNPs. The culture was maintained on Luria Bertani (LB, HiMedia, Mumbai, India) agar plate/slant at 4 °C and as glycerol stocks at –80 °C. For each experiment, loopful of the culture was inoculated in LB broth and incubated overnight at 37 °C/150 rpm to obtain logarithmic phase bacteria.

2.3. Antibacterial Activity of AgNPs Against

A. baumannii AIIMS 7

The minimum inhibition concentration (MIC) and minimum bactericidal concentration (MBC) was determined

by broth micro-dilution method using 96-well microtitre plates as per the CLSI standard guidelines.⁴⁴ Two-fold serial dilutions of AgNPs and antibiotics were made using Mueller Hinton (MH) broth (HiMedia). To each well, 5 µl inoculum ($\sim 5 \times 10^5$ CFU/ml) was added. Each plate included growth and sterility control wells. The microtitre plates were incubated at 37 °C for 20 h and results were recorded. The lowest concentration completely inhibiting the growth as detected by unaided eye was reported as MIC. Aliquot of 5 µl from the wells was taken and spotted onto MH agar plates to determine MBC. The assays were performed in duplicate.

2.4. Fractional Inhibitory Concentration Index (FIC_i)

To determine the effect of AgNPs on activity of antibiotics, FIC_i for combination of AgNPs and antibiotics (ampicillin, amoxicillin, doxycycline, tetracycline and erythromycin) was calculated. FIC_i was determined using MH broth through checkerboard micro titration method in a 96-well microtitre plate, which employs testing of various combinations of two antimicrobials, highest being the MIC of both.^{45,46} The plates were incubated at 37 °C for 20 h and results were recorded visually by naked eye as growth/no growth. The FIC_i was calculated as follows—

$$FIC_A = MIC_A \text{ in combination} / MIC_A$$

$$FIC_B = MIC_B \text{ in combination} / MIC_B$$

where, A = AgNPs and B = antibiotics

$$FIC_i = FIC_A + FIC_B$$

The interaction was defined as synergistic if the FIC_i was ≤ 0.5 , as additive if the FIC_i was >0.5 to 1.0, as indifferent if the FIC_i was >1.0 to 2.0 and as antagonistic if the FIC_i was >2.0 .

2.5. Biofilm Formation

Biofilm formation in *A. baumannii* AIIMS 7 was determined using crystal violet assay. Briefly, 5 µl of OD ($=0.5$ McFarland) adjusted culture was added to 195 µl of LB in a 96-well microtitre plate followed by incubation at 37 °C for 6, 12, 18, 24 and 30 h. After incubation, media was carefully removed and wells were washed thrice with 200 µl of phosphate buffered saline and allowed to dry. Wells were then stained with 0.1% crystal violet (HiMedia) for 15 min. The stain was removed by inverting and tapping and again the wells were kept for drying. The stain was solubilize by addition of 200 µl ethanol followed by reading absorbance at 570 nm in spectrophotometer. Suitable controls were also maintained. Experiment was performed in triplicate and the quantitative value was expressed as the average $\pm$ standard deviation.

2.6. Minimum Biofilm Eradication Concentration (MBEC)

MBEC assay was performed using MBEC™ Biofilm Inoculators (formerly Calgary biofilm device) with 96-well base, which consist of a plastic lid with 96 pegs and a corresponding base of 96-wells. Biofilms are established on pegs and are transferred to new 96-well plate for MBEC determination. Briefly, overnight grown OD adjusted culture of *A. baumannii* AIIMS 7 was added to the wells. Plate was covered with the MBEC™ lid and incubated for 18 h at 37 °C for biofilm formation. After incubation, pegs were rinsed with sterile saline for 1–2 min and exposed to a challenge plate containing serial two-fold dilution gradient of antibiotics (0.1–500 mg/ml) and AgNPs (0.1–4 mg/ml) for 24 h at 37 °C. To evaluate the synergistic effect of AgNPs, a challenge plate having combination of both AgNPs and antibiotics was also tested, where sub-inhibitory concentration of AgNPs was employed. After the challenge, the MBEC™ lid was transferred to a recovery plate having LB broth. The recovery plate was sonicated for 30 min to dislodge the biofilm. The MBEC™ lid was discarded. Recovery plate, covered with sterile non-pegged lid, was further incubated for 20 h at 37 °C and observed for turbidity in the wells. MBEC was taken as the lowest concentration of antimicrobial eradicating biofilm as evident by clear wells in recovery medium.

2.7. Visualization Under FEG-SEM

The pegs, with 18 h preformed biofilm of *A. baumannii* AIIMS 7, were treated with sub-inhibitory concentration of antibiotic, AgNPs and combination of both for 16 h. These were fixed in 4% glutaraldehyde (prepared in 0.1 M PBS, pH 7.4) for 30 min at room temperature followed by sequential dehydration in 50%, 70%, 90% and absolute ethanol for 5 min each. The pegs were then viewed under FEG-SEM (JEOL JSM-7600F, Japan).

2.8. Effect of AgNPs on Bacterial Growth

The effect of nanoparticles on growth of *A. baumannii* AIIMS 7 was estimated by plotting growth curve on exposure with different concentrations of AgNPs. Two-fold serial dilutions viz. 8, 4, 2, 1, 0.5 and 0.25 µg/ml, of AgNPs was prepared in LB broth in a 24-well microtitre plate. OD adjusted ($\sim 5 \times 10^5$ CFU/ml) culture was added to each well. Positive (media and culture), negative (media and AgNPs) and sterility (media) controls were also maintained. The plate was incubated at 37 °C/150 rpm. OD was taken at regular intervals of 1 h at 600 nm using a spectrophotometer. The experiment was done in triplicate and the values were expressed as mean $\pm$ standard deviation.

2.9. Effect on Cell Viability

Cell viability was quantitatively measured by using Live/Dead® BacLight™ Bacterial Viability Kit (Invitrogen) using SpectraMax M5 Multi-mode Microplate Reader

(Molecular Devices LLC, Sunnyvale, CA, USA). The protocol involves determination of cell viability from a standard plot. To prepare the standard graph, OD adjusted ($\sim 1 \times 10^8$ CFU/ml) *A. baumannii* AIIMS 7 suspension (live and killed) were mixed to achieve live:dead cells ratio of 0:100, 10:90, 30:70, 50:50, 70:30, 90:10 and 100:0 in a final volume of 2 ml. An aliquot of 100 µl from each tube was dispensed into separate wells of an opaque 96-well flat-bottom microtitre plate followed by addition of 100 µl of 2 $\times$ stain prepared by mixing SYTO 9 and propidium iodide (PI) dyes. The plate was further incubated at room temperature in dark for 15 min. Fluorescence was measured for SYTO 9 (green) at excitation and emission wavelength of 485 and 530 nm, respectively. For PI (red), fluorescence was measured at excitation wavelength of 485 nm and emission wavelength of 630 nm. Fluorescence data was analyzed as follows–

$$\text{Ratio}_{G/R} = \frac{\text{Fluorescence intensity}_{(\text{SYTO } 9)}}{\text{Fluorescence intensity}_{(\text{PI})}}$$

The graph of Ratio_{G/R} versus percentage of live cells was plotted. From this standard plot, percentage of live cells in bacterial culture treated with 3, 10 and 30 µg/ml AgNPs can be deduced by measuring the fluorescence intensity. Appropriate controls of AgNPs were taken. The experiment was performed in triplicate. The values are expressed as mean $\pm$ standard deviation.

2.10. Endogenous ROS Production

OD adjusted culture of *A. baumannii* AIIMS 7 ($\sim 1 \times 10^8$ CFU/ml) was treated with 10 µg/ml AgNPs for 2 h. Cells were harvested, washed twice with phosphate buffer and resuspended in the same buffer. The probe 2',7'-dichlorofluorescein diacetate (DCFDA; HiMedia) was then added to the suspension at final concentration of 10 µM. Oxidation of DCFDA in by ROS produces 2',7' dichlorodihydrofluorescein (DCF), which is highly fluorescent. The fluorescence intensity (FI) was acquired at regular intervals of 30 min for 2 h using spectrophotometer at excitation wavelength of 485 nm and emission wavelength of 538 nm. ROS generation activity of AgNPs was also evaluated in presence of 10 mM concentration of glutathione and L-ascorbic acid. To determine the ROS variation, the FI of control were subtracted from those of cell treated with compounds. Also, the FI was normalized in relationship to OD₆₀₀ of each sample. The experiment was performed in triplicate. The values are expressed as mean $\pm$ standard deviation.

2.11. Interaction with Free Thiol Groups

The reduction of Ellman reagent (DTNB, 5',5'-dithiobis-(2-nitrobenzoic acid); HiMedia) by free thiol groups produces 5-thio-2-nitrobenzoic acid, which can be monitored spectrophotometrically at 412 nm. Increasing amount of

AgNPs (5, 10, 25 and 50 $\mu\text{g/ml}$) were mixed with 500 μM glutathione and L-cysteine in phosphate buffer (pH 7). After 1 h, 500 μM DTNB was added to the suspension and incubated for 10 minutes. Absorbance was measured at 412 nm. Suitable controls were also maintained. The experiment was performed in triplicate. The values are expressed as mean $\pm$ standard deviation.

2.12. Visualization Under SEM

OD adjusted culture of *A. baumannii* AIIMS 7 ($\sim 1 \times 10^8$ CFU/ml) was treated with 10 $\mu\text{g/ml}$ AgNPs for 6 h. The cells were harvested and washed followed by fixation in 4% glutaraldehyde and sequential dehydration. After drying the cells on a piece of glass surface, the sample was observed under FEGSEM (JEOL JSM-7600F, Japan). A culture with no AgNP treatment was kept as control.

3. RESULTS AND DISCUSSION

3.1. Antibacterial Activity of AgNPs

The broth micro-dilution assay was performed to determine MIC of AgNPs and antibiotics against *A. baumannii* AIIMS 7 (Table I). For AgNPs, it was 16 $\mu\text{g/ml}$. Owing to high surface area-to-volume ratio at nano-size, AgNPs are able to inhibit the bacteria at such a low concentration.⁴⁷ β -lactam antibiotics, amoxicillin and ampicillin, showed very high MIC of 2048 and 4096 $\mu\text{g/ml}$, as also reported by Singh et al.⁴³ Comparatively lower MIC of 64 $\mu\text{g/ml}$ was obtained for erythromycin. Tetracycline class of antibiotics i.e., doxycycline and tetracycline, were found to be more active to inhibit *A. baumannii*, where the MIC was in the range of 1 to 4 $\mu\text{g/ml}$. On comparing the results with CLSI standards (2012) of MIC breakpoints, it was found that *A. baumannii* AIIMS 7 exhibited resistance to ampicillin, amoxicillin and erythromycin.⁴⁴ Our results are in accordance with the study of Ghosh et al.,²³ where disc-diffusion assay was performed with 22 antibiotics.²³ Similar pattern was observed for MBC of antibiotics. However, MBC of AgNPs is similar to that of doxycycline and

tetracycline, indicating that AgNPs can sensitize the bacteria immensely and kill them at a concentration equivalent to that of effective drugs. The significance of this study should be considered on the basis that bacteria are rapidly developing resistance to traditionally available antibiotics and there is a prompt need to seek new alternatives. In our earlier study, we have shown that MIC and MBC values of antibiotics reduced in presence of AgNPs; however, exact nature of interaction in the combination was not determined.⁴³

3.2. Fractional Inhibitory Concentration Index (FICi)

Pharmacodynamic interactions are those where the activity of one drug is altered by presence of another drug during co-administration.⁴⁸ The net effect of such combination can be additive, synergistic or inhibitory, which may have diverse effects on bacterial survival.^{49,50} Fractional inhibitory concentration studies were pursued to establish the pharmacodynamic interactions between AgNPs and antibiotics belonging to three different classes (Table I). The results clearly showed synergistic effects of AgNPs with antibiotics belonging to tetracyclines (doxycycline and tetracycline) and macrolide (erythromycin) class. Additive effect was seen with combination of AgNPs with β -lactams (ampicillin and amoxicillin). The rationale behind FICi experiment is based on two possibilities—(a) beneficial effect of certain drugs in combination with AgNPs to provide enhanced antimicrobial activity suitable to handle infections of mixed etiology, and (b) adverse effect of AgNPs on commonly used antibiotics. Jain et al.⁴⁶ also studied the FICi for combination of AgNPs with various antibiotics, where they reported the additive and synergistic activity of AgNPs with streptomycin, kanamycin, polymyxin B sulfate, ampiclox and ceftazidime.⁴⁶ Antagonistic effect was seen with combination of chloramphenicol and AgNPs contrary to our study where only additive and synergistic effects were seen with the tested antibiotics. de Souza et al.⁵¹ studied the combined antimicrobial activity of colloidal silver (silver-water dispersion) and antibiotics against seven bacterial strains and showed the synergistic activity with amikacin and cefoperazone.⁵¹

Table I. Antibacterial activity of AgNPs and their effect on antibiotics.

Antibiotic	MIC	MBC	MIC in combination		FIC _A	FIC _B	FICi	Interaction
			MIC _A	MIC _B				
AgNPs	16	64						
Ampicillin	4096*	4096*	8	32	0.5	0.008	0.508	Additive
Amoxicillin	2048	2048	8	32	0.5	0.016	0.516	Additive
Doxycycline	1	64	0.125	0.125	0.008	0.125	0.133	Synergistic
Tetracycline	4*	64*	0.5	1	0.032	0.25	0.266	Synergistic
Erythromycin	64	256	0.0625	15.6	0.004	0.243	0.247	Synergistic

Notes: The values (*) are taken from our earlier study.⁴³ The assays were performed against *A. baumannii* AIIMS 7. MIC and MBC values are expressed as $\mu\text{g/ml}$. MIC_A is MIC of AgNPs when exposed to combination of AgNPs and antibiotics. MIC_B is MIC of antibiotics when exposed to combination of AgNPs and antibiotics. FIC_A and FIC_B are FIC of AgNPs and antibiotics, respectively.

Abbreviations: AgNPs, silver nanoparticles; MIC, minimum inhibitory concentration; MBC, minimum bactericidal concentration; FIC, fractional inhibitory concentration; FICi, fractional inhibitory concentration index.

Unlike antibiotics, AgNPs are known to exhibit antimicrobial activity through multiple mechanisms, such as morphology distortion, oxidative stress and inhibition of DNA and protein synthesis.^{35–37} It is important to note that our AgNPs showed pronounced activity with tetracyclines and macrolides. It may be because these antibiotics target protein synthesis machinery,^{52–54} affecting the overall functionality and survival of bacteria, when co-administered with AgNPs. Moreover, *Acinetobacter* sp. produce β -lactamase enzymes which may confer resistance to β -lactam antibiotics viz. ampicillin and amoxicillin;⁵⁵ hence, comparatively lower effect is seen with combination of these antibiotics and AgNPs. It is quite possible that the positive or additive effect with such resistant antibiotics is prominently due to presence of AgNPs with multi-mode bactericidal machinery. This study suggested that AgNPs can be conjugated to antibiotics having less or no activity, which usually require high concentration to kill pathogenic bacteria, to develop effective formulations in future.

3.3. Biofilm Formation by *A. baumannii* AIIMS 7

Colonization capacity of *A. baumannii* AIIMS 7 was detected by crystal violet assay. It is clear from Figure 1 that the bacterial cells attached to the wells of microtitre plate and continued to mature till 18 h after which the biofilm detached from the surface. The pattern can be correlated with the life cycle of biofilm in bacteria, which majorly consist of three stages, viz. attachment, growth/maturation and detachment.⁵⁶ The similar pattern of biofilm formation has also been observed in clinical *A. haemolyticus* MMC 8; however, the bacteria form maximum biofilm in 72 h in microtitre well.⁵⁷ This indicates that *A. baumannii* is more potent biofilm producer and matures rapidly in biofilm mode, compared to *A. haemolyticus*. This render them strength to survive in extreme conditions, such as high pH, dessication, ultraviolet light and high dose of drugs.^{15–18}

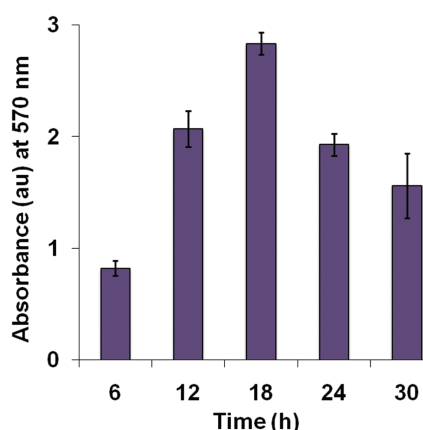


Figure 1. Biofilm formation by *A. baumannii* AIIMS 7 at various time points.

3.4. Minimum Biofilm Eradication Concentration (MBEC)

Preformed 18 h biofilm of *A. baumannii* AIIMS 7 was subjected to treatment with various concentrations of AgNPs and antibiotics (ampicillin from β -lactams, doxycycline from tetracyclines and erythromycin from macrolides) for 24 h to determine the MBEC. MBEC of AgNPs was 2 mg/ml and that of ampicillin, erythromycin and doxycycline was 256, 128 and 1 mg/ml, respectively (Table II). Gaidhani et al.³³ also reported the biofilm disruption activity of AgNPs against clinical strains of *Acinetobacter* at 2 mg/ml concentration.³³ We also tested the effect of presence of sub-inhibitory concentration of AgNPs (1 mg/ml) on MBEC values of antibiotics. Erythromycin showed a significant reduction in MBEC value, from 128 to 4 mg/ml, in presence of AgNPs. Two-fold reduction was observed for the combination of doxycycline and AgNPs, where MBEC value was decreased to 0.5 mg/ml. However, ampicillin did not exhibit any change in presence of nanoparticles. In a similar study, Hwang et al.⁵⁸ reported the enhanced biofilm inhibition with combination of chemically synthesized nano-Ag and conventional antibiotics, ampicillin, chloramphenicol and kanamycin.⁵⁸ However, this is the first study to describe the effect of co-administration of biological AgNPs (synthesized using bacteria) and antibiotics. As mentioned in the Section 3.2, the biofilm disruption effect of combination of AgNPs with antibiotics can be correlated with FIC_i, which explains the synergistic activity of combination of AgNPs with erythromycin and doxycycline, and indifferent activity with ampicillin.

It is evident from Tables I and II that *A. baumannii* AIIMS 7 require very high ($>1000 \times$ MIC) concentration of antibiotics for complete eradication of biofilms confirming the fact that they offer greater resistance in biofilm mode as compared to planktonic form. However, comparatively lower concentration of AgNPs (2 mg/ml) can eradicate the biofilm completely. Many reports are available demonstrating antibiofilm action of AgNPs in Gram positive and Gram negative bacteria.^{34, 59–61} The biofilms possess thick layers of EPS forming an outer protective covering to restrict the entry of antibiotics due to which absolute eradication of biofilms is not achieved.

Table II. MBEC (mg/ml) of antibiotics in presence and absence of AgNPs.

Antibiotic	MBEC (without AgNPs)	MBEC (with AgNPs)
AgNPs	–	2
Ampicillin	256	256
Doxycycline	1	0.5
Erythromycin	128	4

Notes: MBEC was determined using MBEC™ Biofilm Inoculators with preformed 18 h biofilm of *A. baumannii* AIIMS 7. Concentration of AgNPs used in combination with antibiotics was 1 mg/ml.

Abbreviations: AgNPs, silver nanoparticles; MBEC, minimum biofilm eradication concentration.

However, owing to nano size, AgNPs are capable to penetrate and destroy this EPS matrix.³³ Due to high surface area-to-volume ratio, the nanoparticles are able to exert the biofilm disruption activity in low concentration. This initial sensitization of biofilms via AgNPs paves way for antibiotics to act and disrupt them during co-administration. Besides this, AgNPs have been reported to reduce the microbial biomass, number of surviving cells and production of exopolysaccharides and proteins in biofilms.³⁴

3.5. Effect on Biofilm Structure

Electron micrographs revealed the confluent lawns in untreated biofilm wherein the bacterial cells, having smooth surface, were present in intimate contact with each other (Fig. 2(a)). Although treatment with drug erythromycin caused biofilm disruption, few cells with intact smooth-surfaced morphology were also observed (Fig. 2(b)). However, treatment with AgNPs and combination of AgNPs and erythromycin led to significant biofilm eradication and loss of confluent growth (Figs. 2(c, d)). Besides this, presence of discrete bacterial cells with no contact and severe morphological disruption was also observed. It is important to note that for FEG-SEM analyses, sub-inhibitory concentration of AgNPs, antibiotic

and combination has been employed so that disruption of biofilm can be visualized. The pegs were devoid of any adhered cells when treated with MBEC values of AgNPs and combination indicating absolute eradication.

Acinetobacter biofilms present a major challenge due to their ubiquitous nature, ability to survive in diverse ecological niches, colonization and biofilm-forming capacity of medical devices, such as catheters, implants, prosthetics, cardiac valves and artificial joints.^{14, 62} The biofilms cause severe illness and diseases in immuno-compromised patients, especially in case of urinary and respiratory tract infections, ocular infection, otitis media, endocarditis, pneumonia, septicemia, bacteremia, and necrotizing fasciitis etc.^{63–65} Mortality can be attributed to drug resistance, ineffective empirical therapy and unavailability of therapeutic options.²² It has been suggested that nanoparticles cause inefficient surface attachment, membrane disintegration, efflux of cellular materials and morphological distortion of cells present in biofilms.²² However, treatment with combination of AgNPs and antibiotic resulted in complete biofilm eradication at comparatively lower concentration of antibiotic and in lesser time. Hereby, the observed biofilm disruption activity of combination of nano silver and antibiotics against *A. baumannii* pathogen is an immensely significant perspective,

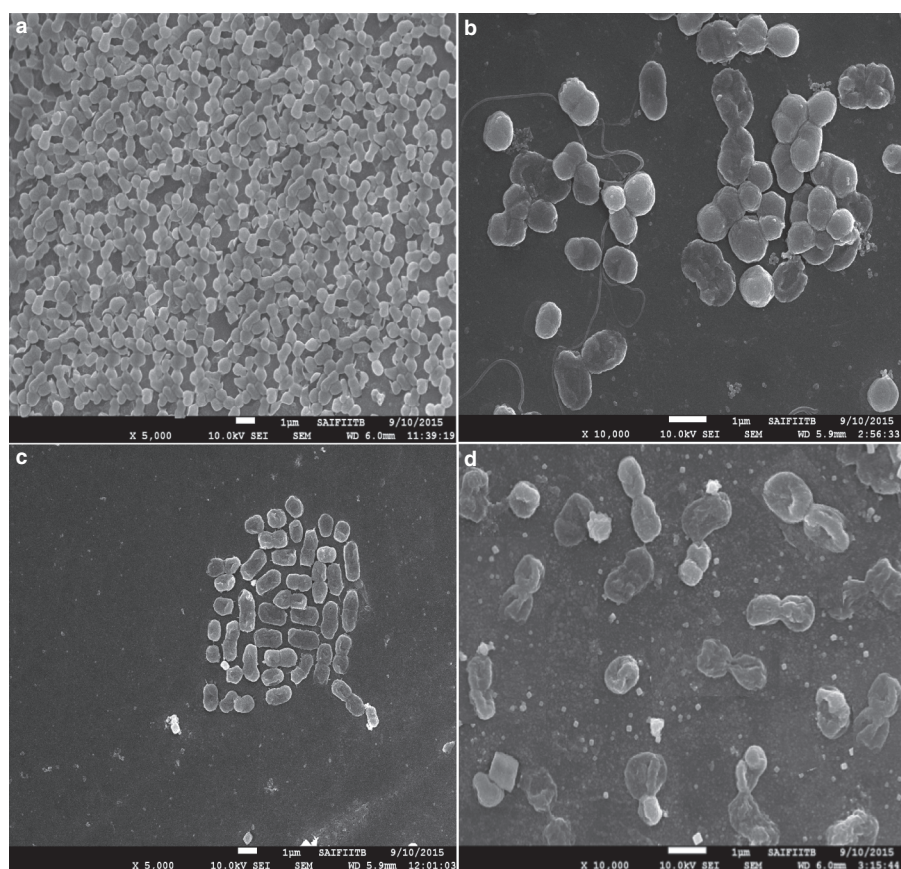


Figure 2. FEG-SEM images of biofilm disruption in *A. baumannii* AIIMS 7 after 16 h treatment. (a) control, (b) erythromycin, (c) AgNPs, and (d) erythromycin and AgNPs. (scale bar represents 1 μm).

which should be further investigated for designing and development of effective therapeutic strategy against these MDR pathogenic biofilms of *A. baumannii*.

3.6. Effect of AgNPs on Bacterial Growth

The growth inhibition effect of AgNPs on *A. baumannii* AIIMS 7 was determined through growth curve analyses. Figure 3 shows that the bacterial growth was dependent upon the AgNP concentration. In absence of AgNPs, the bacteria reached its exponential phase in 3 h. However, a concentration-dependent increase in the lag phase was observed in presence of AgNPs. Exposure to sub-inhibitory concentration of 8 $\mu\text{g/ml}$ AgNPs, lengthened the lag phase of bacteria from 2 h to 20 h. Even the lowest concentration of 0.25 $\mu\text{g/ml}$ affected its growth rate, though not significantly, and the bacterial cells remained in the lag phase for 3 h. Similar observations have been reported by Zhang and colleagues (2014) where they reported a delay in growth of *Escherichia coli*, *Pseudomonas aeruginosa* and *Staphylococcus aureus* in presence of bio-AgNPs.³⁴

3.7. Effect of AgNPs on Cell Viability

Exposure of grown cells of *A. baumannii* AIIMS 7 to increasing concentration of AgNPs in planktonic state resulted in decrease in cell viability as estimated by spectrofluorometer. At 30 $\mu\text{g/ml}$ AgNP concentration, death in 50% of the cells was achieved in first 6 h and merely 5% cells were viable after 24 h, as shown in Figure 4. However, at lower concentrations of AgNPs relatively higher viability was seen at the end of the treatment period. These observations are in accordance with the report of Zhang et al.³⁴ They reported 44–53% viability in *P. aeruginosa* cells after exposure to 7.8–62.5 $\mu\text{g/ml}$ AgNPs, which indicates that the AgNPs synthesized by cell-free extract of *A. calcoaceticus* are comparatively more efficient.

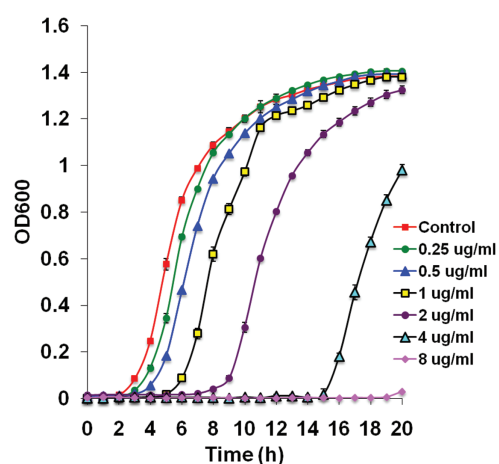


Figure 3. Effect of AgNPs on growth of *A. baumannii* AIIMS 7 in presence of various concentration of AgNPs.

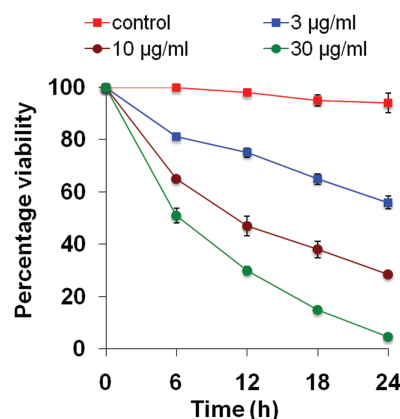


Figure 4. Effect of AgNPs on *A. baumannii* AIIMS 7 viability.

3.8. Endogenous ROS Production

It has been proposed that AgNPs may produce ROS, including superoxide radicals, hydroxyl radicals, hydrogen peroxide, etc., which destroys cellular proteins and DNA leading to oxidative stress in the cells.⁶⁶ To determine the level of endogenous ROS formed in *A. baumannii* AIIMS 7 cells treated with AgNPs, the probe DCFDA was added and production of DCF was monitored by measuring fluorescence intensity (FI). Figure 5 revealed the time-dependent increase in FI in cells treated with AgNPs for up to 60 min indicating the ability of AgNPs to generate ROS intracellularly. The fluorescent dye DCF is stable for 45 to 60 min due to which a decrease in FI was observed at 90 min. However, treatment with AgNPs in presence of known antioxidants, such as ascorbic acid, glutathione and L-cysteine, resulted in decreased production of endogenous ROS. It is worth noting that ROS production was significantly inhibited in presence of glutathione and L-cysteine as compared to ascorbic acid. This could be due to the interaction of AgNPs with glutathione and L-cysteine. Till date no study has been pursued to describe the endogenous ROS production by bacteriogenic AgNPs. However, exposure to chemically synthesized nano-Ags

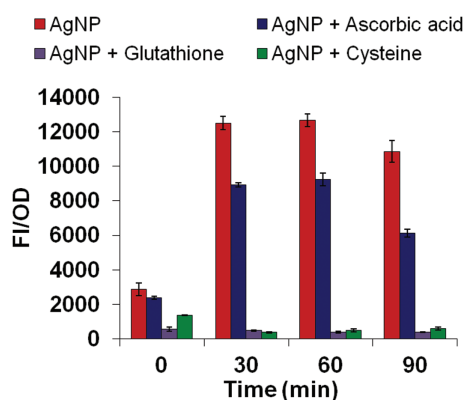


Figure 5. Endogenous ROS production in *A. baumannii* AIIMS 7 in presence of AgNPs.

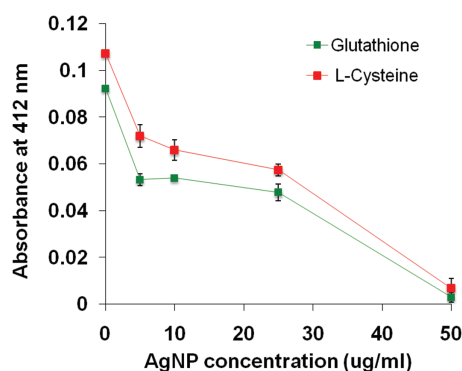


Figure 6. Interaction of AgNPs with free thiol groups of glutathione and cysteine.

has been reported to produce hydroxyl radicals in *Enterococcus faecium*, *S. aureus*, *Streptococcus mutans*, *E. coli* and *P. aeruginosa*.⁵⁸ Tavres et al.⁶⁷ also employed the similar method to elucidate the endogenous ROS production in *E. coli* by carbon-monoxide releasing molecules.⁶⁷

3.9. Interaction with Free Thiol Groups

To further analyze the significant decrease in ROS production by AgNPs in presence of glutathione and L-cysteine, DTNB reduction assay was carried out to examine the possible interaction between AgNPs and free thiol groups present in both glutathione and L-cysteine. A decrease in absorbance with increase in concentration of AgNPs, as shown in Figure 6, indicated that AgNPs interacted with free thiol groups present in glutathione and L-cysteine, which resulted in availability of fewer free thiol groups for reduction of DTNB. This study points at the possibility that AgNPs can interact with intracellular proteins owing to presence of thiol-containing amino acids, like cysteine, as also suggested by Li et al.³⁵ AgNPs have also been shown to possess tendency to attach to sulfur and phosphorus groups that are part of phospholipid cell membrane, membrane proteins and DNA.^{68–70} These interactions may lead to protein degradation or inactivation, which warrants further confirmatory investigation. It will also be interesting to study whether AgNPs have the ability to break the disulfide bonds present in proteins to cause denaturation.

3.10. Effect on Cell Morphology

SEM was used to evaluate the surface morphology of untreated and treated planktonic cells of *A. baumannii* AIIMS 7. Surface of untreated cells was smooth and characteristically coccobacillary, as shown in Figure 7(a). However, significant morphological changes were observed in cells treated with AgNPs (Fig. 7(b)). The cells were rough, flattened and mis-shaped. Elongation of cells was also observed after AgNP treatment. These results are in agreement with the previous studies reporting the serious morphological damage, cell flattening and formation of holes in bacterial cell wall after treatment

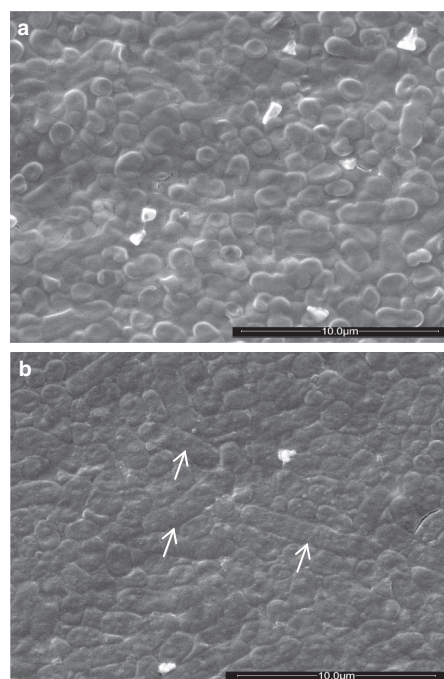


Figure 7. Cellular morphology of *A. baumannii* AIIMS 7 after 6 h exposure to sub-inhibitory concentration of AgNPs. (a) control, and (b) AgNPs treated. Arrows indicate morphology distortion and cell elongation after AgNP treatment. (scale bar represents 10 μ m).

with AgNPs.^{34, 37} Kanmani and Lim⁵⁹ also observed the disruption of bacterial cell membrane and intracellular localization of AgNPs by TEM analyses.⁵⁹ Electrostatic interactions between AgNPs present in solution and the bacterial cell membrane may provide the favourable conditions to nanoparticles to attach on to the cell surface.^{37, 71, 72} This affects the cellular permeability, respiratory functions and electron transport chain.^{73, 74}

4. CONCLUSION

This is the first extensive report to demonstrate antibacterial activity of bacteriogenic AgNPs against a nosocomial pathogen, *A. baumannii*, and their mechanism of action. The nanoparticles exhibited significant antibacterial activity against *A. baumannii* AIIMS 7 in suspension and biofilm mode. They mount a synergistic relationship with doxycycline, tetracycline and erythromycin. These kill the bacteria by disrupting the cell wall and producing ROS. However, one should consider that the way to kill bacteria is highly specific to the respective bacterial strains in terms of its cell wall composition, growth rate, biofilm formation capability and type of nanoparticles. Different synthesizing procedures, size and shape can render AgNPs with variable antimicrobial properties. Moreover, binding of nanoparticles to any microorganism depends on the interaction of the available surface area, thus, small sized AgNPs with larger surface area would

exhibit greater microbicidal activity compared to larger AgNPs.

Disclosure

The author reports no conflicts of interest in this work.

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